ABSTRACT

The rapid growth of social media platforms has significantly increased the spread of misinformation, making fake news detection a critical challenge. Twitter, being one of the most widely used platforms for real-time information sharing, is particularly vulnerable to the انتشار of misleading and false content. This project focuses on developing an automated fake news detection system using machine learning and Natural Language Processing (NLP) techniques. The system analyzes tweet content, linguistic patterns, and user behavior to classify information as real or fake. Text preprocessing methods such as tokenization, stop-word removal, and stemming are applied to clean the data, followed by feature extraction techniques like TF-IDF. Various classification algorithms, including Logistic Regression, Naïve Bayes, and Support Vector Machines, are used to improve prediction accuracy. The model is trained and tested on labeled Twitter datasets. The results demonstrate that the system can effectively identify fake tweets, helping to reduce the spread of misinformation and improve the reliability of online information.

Keywords:
Fake News Detection, Twitter, Natural Language Processing (NLP), Machine Learning, Text Classification, TF-IDF, Logistic Regression, Naïve Bayes, Support Vector Machine (SVM), Social Media Analysis.

Why is this model better than others?

- Uses a **hybrid approach**, combining multiple techniques for better accuracy
- Applies **NLP methods (TF-IDF)** for effective text understanding
- Uses **Logistic Regression**, which is fast and efficient
- Handles **noisy Twitter data** through proper preprocessing
- Provides **real-time predictions** through a user-friendly system

VARUN TEJ KALLAGANTI
23STUCHH010820